

Webscraping

```
In [5]: import requests
        from bs4 import BeautifulSoup

        # Define a function to hold our scraping logic
        def scrape_quotes():
            print("--- Scraping Page 1 ---")

            # Set the website address (URL) we want to visit
            url = "https://quotes.toscrape.com/"

            # Send a request to the website to download the page content
            response = requests.get(url)

            # Check if the request was successful (200 means OK)
            if response.status_code == 200:
                # Convert the raw website code (HTML) into a readable format
                soup = BeautifulSoup(response.text, 'html.parser')

                # Search the page for all quote text and author names
                quotes = soup.find_all('span', class_='text')
                authors = soup.find_all('small', class_='author')

                # Pair up each quote with its author and loop through them one by one
                for quote, author in zip(quotes, authors):
                    # Print the clean text of the quote and the author's name
                    print(f"Quote: {quote.text}")
                    print(f"Author: {author.text}")
                    print('-----')

            else:
                # detailed error message if the website refuses connection
                print(f"Failed to connect. Status code: {response.status_code}")

        # Run the function to start scraping
        scrape_quotes()
```

--- Scraping Page 1 ---

Quote: "The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking."

Author: Albert Einstein

Quote: "It is our choices, Harry, that show what we truly are, far more than our abilities."

Author: J.K. Rowling

Quote: "There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle."

Author: Albert Einstein

Quote: "The person, be it gentleman or lady, who has not pleasure in a good novel, must be intolerably stupid."

Author: Jane Austen

Quote: "Imperfection is beauty, madness is genius and it's better to be absolutely ridiculous than absolutely boring."

Author: Marilyn Monroe

Quote: "Try not to become a man of success. Rather become a man of value."

Author: Albert Einstein

Quote: "It is better to be hated for what you are than to be loved for what you are not."

Author: André Gide

Quote: "I have not failed. I've just found 10,000 ways that won't work."

Author: Thomas A. Edison

Quote: "A woman is like a tea bag; you never know how strong it is until it's in hot water."

Author: Eleanor Roosevelt

Quote: "A day without sunshine is like, you know, night."

Author: Steve Martin

```
In [2]: # Practice only code here form above. 
import requests
from bs4 import BeautifulSoup

def scrape2_all():
    print("----Webcscrapping Function----")
    url = "https://quotes.toscrape.com/"
    response = requests.get(url)

    if response.status_code == 200:
        soup = BeautifulSoup(response.text, 'html.parser')
        quotes = soup.find_all('span', class_='text')
        authors = soup.find_all('small', class_='author')

        for quote, author in zip(quotes, authors):
            print(f"Quote:{quote.text}")
            print(f"Author:{author.text}")
```

```

        print("-----")
    else:
        print(f"Fail to connect: {response.status_code}")
    scrape2_all()

```

----Webcscrapping Function----

Quote:"The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking."

Author:Albert Einstein

Quote:"It is our choices, Harry, that show what we truly are, far more than our abilities."

Author:J.K. Rowling

Quote:"There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle."

Author:Albert Einstein

Quote:"The person, be it gentleman or lady, who has not pleasure in a good novel, must be intolerably stupid."

Author:Jane Austen

Quote:"Imperfection is beauty, madness is genius and it's better to be absolutely ridiculous than absolutely boring."

Author:Marilyn Monroe

Quote:"Try not to become a man of success. Rather become a man of value."

Author:Albert Einstein

Quote:"It is better to be hated for what you are than to be loved for what you are not."

Author:André Gide

Quote:"I have not failed. I've just found 10,000 ways that won't work."

Author:Thomas A. Edison

Quote:"A woman is like a tea bag; you never know how strong it is until it's in hot water."

Author:Eleanor Roosevelt

Quote:"A day without sunshine is like, you know, night."

Author:Steve Martin

API

In [14]: `import requests`

```

def get_weather():
    print("--- Weather API ---")

    # 1. Define the endpoint (The address of the API server)
    base_url = "http://api.openweathermap.org/data/2.5/weather"

```

```

# 2. Define the settings (The specific data we are asking for)
parameters = {
    "q": "Montreal", # The city name
    "appid": "fae762e393c429a1ca033e3ad20744dc", # Your unique API key
    "units": "metric" # Use Celsius instead of Fahrenheit
}

# 3. Send the request (Python automatically builds the full URL using 'params')
response = requests.get(base_url, params=parameters)

# 4. Check if the connection was successful (200 OK)
if response.status_code == 200:

    # 5. Convert the response text into a Python dictionary
    data = response.json()

    # 6. Extract and print the specific details we want
    print(f"City: {data['name']}")
    print(f"Temp: {data['main']['temp']}°C")
    print(f"Weather: {data['weather'][0]['description']}")

else:
    # Handle errors (like typing a wrong city name)
    print(f"Error: {response.status_code}")

# Run the function
get_weather()

```

```

--- Weather API ---
City: Montreal
Temp: -15.97°C
Weather: few clouds

```

```

In [44]: # Practice Code Weather Api
import requests

def get_weather():
    print("---- Weather API ----")
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    parameters = {"q": "Rome", "appid": "fae762e393c429a1ca033e3ad20744dc", "units": "metric"}

    response = requests.get(base_url, params=parameters)
    if response.status_code == 200:
        data = response.json()
        print(f"City:{data['name']}")
        print(f"Temp:{data['main']['temp']} Celsius")
        print(f"Weather:{data['weather'][0]['description']}")
    else:
        print(f"Fail to connect:{response.status_code}")
get_weather()

```

```

---- Weather API ----
City:Rome
Temp:-2.95 Celsius
Weather:clear sky

```

For Revision Combine Logic

In []: ROOT GOAL: GET DATA FROM INTERNET

```
├─ STEP 1: PREPARATION (Same Library)
│   ├── Scraping: import requests
│   └─ API:      import requests
├─ STEP 2: THE DESTINATION (Different Targets)
│   ├── Scraping: URL = "https://quotes.toscrape.com/"
│   │               (Targeting a Visual Page)
│   └─ API:      URL = ".../data/2.5/weather" + params
│               (Targeting a Data Endpoint)
├─ STEP 3: THE KNOCK (Same Command) 🚨
│   ├── Scraping: response = requests.get(url)
│   └─ API:      response = requests.get(url)
├─ STEP 4: THE CHECK (Same Safety) ✅
│   ├── Scraping: if response.status_code == 200:
│   └─ API:      if response.status_code == 200:
├─ STEP 5: THE TRANSLATION (MAJOR SPLIT) 🔗
│   ├── Scraping: soup = BeautifulSoup(response.text, 'html.parser')
│   │               (Translates messy HTML into a Tree)
│   └─ API:      data = response.json()
│               (Translates raw text into a Python Dictionary)
└─ STEP 6: EXTRACTION (Different Tools)
    ├── Scraping: soup.find_all('span', class_='text')
    │               (Tool: "Search" for tags/classes)
    └─ API:      data['main']['temp']
                (Tool: "Lookup" by key names)
```

In [2]: # Webscraping (29 jan morn 1 Recall)

```
import requests
from bs4 import BeautifulSoup

def scrape3_all():
    print("---- Web Scrapping Function ----")
    url = "https://quotes.toscrape.com/"
    response = requests.get(url)

    if response.status_code == 200:
        soup = BeautifulSoup(response.text, "html.parser")
        quotes = soup.find_all('span', class_='text')
        authors = soup.find_all('small', class_='author')

        for quote, author in zip(quotes, authors):
            print(f"Quote:{quote.text}")
            print(f"Author:{author.text}")
            print(f"-----")
    else:
```

```
print(f"Fail to connect:{response.status_code}")
scrape3_all()
```

---- Web Scrapping Function ----

Quote:"The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking."

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Author:Thomas A. Edison

Quote:"A woman is like a tea bag; you never know how strong it is until it's in hot water."

Author:Eleanor Roosevelt

Quote:"A day without sunshine is like, you know, night."

Author:Steve Martin

In [17]: *# Weather API (29 jan morn 1 Recall)*

```
import requests
```

```
def get_weather():
```

```
    base_url = "https://api.openweathermap.org/data/2.5/weather"
```

```
    parameters = {'q': 'Montreal', 'appid': 'fae762e393c429a1ca033e3ad20744dc', 'units'
```

```
    response = requests.get(base_url, params=parameters)
```

```
    if response.status_code == 200:
```

```
        data = response.json()
```

```
        print(f"City:{data['name']}")
```

```
        print(f"Temperature:{data['main']['temp']} celsius")
```

```
        print(f"Weather:{data['weather'][0]['description']}")
```

```
    else:
        print(f"fail to connect: {response.status_code}")
get_weather()
```

City:Montreal

Temperature:-21.04 celsius

Weather:clear sky

Second Recall - Webscrapping and Api

```
In [2]: import requests
        from bs4 import BeautifulSoup

        def quotes_scrape():
            print("----Webscrapping----")
            url = "https://quotes.toscrape.com/"
            response = requests.get(url)

            if response.status_code == 200:
                soup = BeautifulSoup(response.text, 'html.parser')
                quotes = soup.find_all('span', class_='text')
                author = soup.find_all('small', class_='author')

                for quote, author in zip(quotes, author):
                    print(f"Quote:{quote.text}")
                    print(f"Author:{author.text}")
                    print("-----")
            else:
                print(f"Fail to connect {response.status_code}")
        quotes_scrape()
```

----Webscrapping----

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Author:Eleanor Roosevelt

Quote:"A day without sunshine is like, you know, night."

Author:Steve Martin

In [25]: `import requests`

```
def weather_api():
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    params = {'q': 'Mumbai', 'appid': 'fae762e393c429a1ca033e3ad20744dc', 'units': 'metric'}
    response = requests.get(base_url, params=params)

    if response.status_code == 200:
        data = response.json()
        print(f"City:{data['name']}")
        print(f"Temp:{data['main']['temp']}")
        print(f"Weather:{data['weather'][0]['description']}")
    else:
        print(f"Fail to connect: {response.status_code}")
weather_api()
```


City:Mumbai

Temp:27.99

Weather:haze

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